

An Internship Report
Of
C++ Programming Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)

By

Name of Student: ADITYA PANDEY

Programme: B.Tech CSE, Batch 2024–2028

Institute: IILM University, Greater Noida, U.P.

Internship Duration: 03 August 2026 – 02 September 2026

Certificate ID: THX-AUG0326-247

CANDIDATE'S DECLARATION

I, ADITYA PANDEY, do hereby declare that the internship report titled “C++ Programming Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: ADITYA PANDEY

Programme: B.Tech CSE, Batch 2024–2028

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work.

I would like to thank Thiranex for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of C++ Programming. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: ADITYA PANDEY

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3. INTERNSHIP COMPLETION CERTIFICATE

This is to certify that Aditya Pandey has successfully completed an internship in C++ Programming from 03 August 2026 to 02 September 2026, awarded by Thiranex (Certificate ID: THX-AUG0326-247), signed by Hariharan M, Founder & CEO.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Virtual Internship undertaken under Thiranex, in the domain of “C++ Programming”. The internship was carried out from 03 August 2026 to 02 September 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary aim of this internship was to strengthen practical programming skills using C++, moving from individual language concepts to a complete, working, menu-driven console application. The learning journey emphasized programming fundamentals, object-oriented design, file handling, and the development of a persistent, real-world data-management project.

The internship followed a structured, progressive curriculum combining conceptual learning with hands-on implementation, culminating in the development of a console-based Student Management System and the award of a Certificate of Achievement by Thiranex.

4.2 Organization Profile

Thiranex is an organization working in the area of skill development and future technology, running structured virtual internship programs for engineering students. Thiranex designs industry-aligned, hands-on curricula covering core programming languages and practical software-development skills, delivered through a progressive, project-based learning journey that helps students build real coding capability rather than purely theoretical knowledge.

4.3 Problem Statement

Programming concepts are most valuable when they can be applied to solve a real task. While many engineering students learn the theoretical foundations of C++ during coursework, few get structured, hands-on exposure to combining language fundamentals, object-oriented programming, and file handling into a single, working application. The problem addressed by this internship was to practically learn and apply the complete C++ development workflow — from core syntax and logic, to designing classes and functions, to managing persistent data, to building a fully functional console application.

4.4 Project Objectives

- To strengthen C++ fundamentals, including syntax, control flow, arrays, strings and functions.
- To apply classes, objects and encapsulation in application design (Object-Oriented Programming).
- To store and retrieve records using persistent, file-based storage.
- To implement add, display, search, update and delete (CRUD) functionality.
- To handle user input carefully and improve application reliability through validation.
- To combine all concepts into a complete, menu-driven console application.

4.5 Scope of the Project

The scope of this internship spans the complete C++ application-development workflow — starting from programming foundations (variables, data types, operators, conditions, loops, arrays, strings and functions), moving through object-oriented programming (classes, objects, encapsulation, constructors and member functions), and extending into practical application development involving file handling, CRUD operations, input validation and a menu-driven interface. The internship was limited to a virtual, project-based learning environment provided by Thiranex, and culminated in a console-based Student Management System rather than deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Programming Language: C++
- Programming Paradigm: Object-Oriented Programming (OOP) — classes, objects and encapsulation
- Data Persistence: File handling for persistent record storage
- Libraries: Standard Template Library (STL) / Strings for common data and text operations
- Development Environment: Visual Studio Code
- Compiler: G++ (compiler for building C++ programs)
- Input Validation: Regex, used where required
- Application Type: Console applications with a menu-driven interface

4.7 System Architecture

The architecture followed during the internship, from user input to persistent student records, can be summarized in the following layers:

- User Interface Layer: A menu-driven console input drives all interaction with the application.
- Application Logic Layer: Functions, validation routines and control flow process each user request.
- Data Management Layer: Classes and objects encapsulate student data and implement CRUD operations.
- File Layer: Dedicated routines read and write student records to persistent storage.
- Persistent Storage Layer: All records are stored in a text-based file, studentRecord.txt.

4.8 Methodology

The internship followed a structured, eight-stage methodology combining conceptual learning with hands-on implementation, culminating in a complete, tested console application, as summarized in the table below.

Stage	Module	Description
1	C++ Foundations	Syntax, variables, operators and control flow.

2	Functions & Arrays	Modular programming, arrays and strings.
3	OOP	Classes, objects, constructors and encapsulation.
4	File Handling	Reading, writing and persistent storage.
5	CRUD Design	Add, search, update and delete operations.
6	Validation	Input handling and error management.
7	Integration	Combining modules into one application.
8	Testing & Documentation	Debugging, refinement and project presentation.

Note: There was no stipend associated with this internship.

Each stage concluded with practical implementation of the corresponding concepts. In the final phase, the individual modules were integrated, tested and documented, resulting in a complete Student Management System and the award of a Certificate of Achievement.

4.9 Expected Outcomes

- Ability to write structured C++ programs using core language features.
- Practical understanding of Object-Oriented Programming and modular design.
- Hands-on experience with file handling and persistent record storage.
- Capability to implement CRUD operations in a menu-driven application.
- Improved approach to input validation, debugging and error handling.
- Experience combining multiple programming concepts into one working project, resulting in stronger C++ fundamentals and practical console-application development skill.

4.10 Certificate of Completion

The Certificate of Achievement issued by Thiranex, confirming successful completion of the C++ Programming Internship from 03 August 2026 to 02 September 2026 (Certificate ID: THX-AUG0326-247), is attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

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